Calibrate Where You Deploy: Conformal Sensor Escalation Under Operating-Point Shift

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[TODO: e-mail, coautores, e o agradecimento de fomento]

***Abstract*—Condition-monitoring nodes must decide, window by window, whether the cheap always-on sensor suffices or the expensive acquisition chain has to be woken. That decision is normally driven by a softmax threshold τ with no operational meaning. We replace it with a split-conformal gate: escalate iff the prediction set is not a singleton. The risk level α then becomes a specification, and the escalation rate — hence the energy bill — becomes a *measured consequence* rather than a second tuned knob. On the KAIST bearing dataset (45 recordings, 5 classes, 3 torque loads) we report three results. First, at matched escalation rate the conformal gate is statistically indistinguishable from the softmax threshold in accuracy (Δ macro-F1 $= -0.003 \pm 0.015$); its value is specification, not accuracy. Second, the coverage guarantee holds under recording-grouped validation ($+0.046$) but collapses under load shift (-0.146 mean, -0.453 worst case: a nominal 80% delivers 55 %). Third, *five* recordings from the target operating point restore nominal coverage — and discarding the source calibration set beats pooling it.**

***Index Terms*—conformal prediction, condition monitoring, multimodal sensing, TinyML, distribution shift**

I. Introduction

Battery-powered condition-monitoring nodes are dominated, energetically, by their analog front ends. A MEMS accelerometer sampled at 25.6 kHz together with its conditioning chain and high-rate ADC costs orders of magnitude more than a current transformer sampled slowly, and both cost more than the microcontroller cycles spent classifying the resulting window. The standard response is a *cascade*: run a cheap always-on model, and wake the expensive chain only when the cheap model looks uncertain [1].

The trouble is the wake-up rule. In practice it is a threshold on the softmax output, $\max_y \hat{p}(y \mid x) < \tau$, with τ swept until the accuracy–energy trade-off looks acceptable. This is unsatisfying for two reasons. Softmax scores of small discriminative models are not calibrated probabilities, so τ has no units and no operational meaning; and because τ is tuned, the escalation rate — the quantity that actually sets battery life — is an outcome of tuning rather than something the integrator specifies.

Split conformal prediction [2]–[5] offers a different handle. Given a held-out calibration set and a target risk α, it returns prediction *sets* with finite-sample marginal coverage $\mathbb{P}(Y \in C(X)) \geq 1 - \alpha$ under exchangeability, with no assumption on the underlying model. We use the set itself as the gate: escalate iff $|C(x)| \neq 1$. The knob is now α, a risk the integrator states, and the escalation rate is measured.

Our contributions are:

- We formulate sensor escalation as a conformal set-size test and show that, at matched escalation rate, it neither helps nor hurts accuracy relative to a tuned softmax threshold (Table II). We report this as a negative result: the contribution is the change of knob, not an accuracy gain.
- We show that the conformal guarantee, which holds under recording-grouped cross-validation, *collapses* when the machine moves to a torque load absent from calibration, and that it collapses unevenly across operating points (Fig. 1).
- We quantify the fix: five recordings from the target operating point restore nominal coverage, and *discarding* the stale calibration set is roughly twice as good as pooling it (Table III, Fig. 2).
- As a by-product, we map the sensor-chain frontier and find that nine time-domain features from a *single* accelerometer come within 0.06 macro-F1 of a four-accelerometer chain with FFT features, at $18\times$ lower acquisition cost (Table I).

[TODO: parágrafo de related work. Cobrir (i) cascatas e wake-up hierárquico de sensores, (ii) conformal prediction em PHM — *a checagem de prior art ainda não foi feita, ver a nota no fim do arquivo*, e (iii) a prior art de transferência cross-modal vibração→corrente [6], que precisa aparecer explicitamente para não parecer que a ignoramos]

II. Conformal Escalation

A. Split conformal prediction

Let $\hat{p}(\cdot \mid x)$ be any trained classifier over K classes and let $\mathcal{D}_{\text{cal}} = \{(x_i, y_i)\}_{i=1}^{n}$ be a calibration set disjoint from training. Fix a nonconformity score $s(x, y)$. We use two: the *least-ambiguous* (LAC) score [7]

$$s_{\text{LAC}}(x, y) = 1 - \hat{p}(y \mid x), \tag{1}$$

which yields the smallest average set size at a given coverage, and the *adaptive prediction set* (APS) score [8], which accumulates sorted class probabilities down to the true class and trades set size for conditional coverage.

Let $\hat{q}$ be the $\lceil (n+1)(1-\alpha) \rceil / n$ empirical quantile of $\{s(x_i, y_i)\}$. The prediction set at a new x is

$$C(x) = \{\, y : s(x, y) \le \hat{q} \,\}. \tag{2}$$

If (x_i, y_i) and the test point are exchangeable, then $\mathbb{P}(Y \in C(X)) \ge 1 - \alpha$ for any finite n, with no distributional or model assumption [3], [4]. The inflated quantile level is what makes the bound finite-sample; when $\lceil (n+1)(1-\alpha) \rceil / n > 1$ the calibration set is too small for the requested α and the set degenerates to all classes, which we report rather than hide.

B. The gate

The cheap model runs on every window. We escalate — wake the expensive chain and let its model decide — exactly when the cheap model's conformal set is not a singleton:

$$\text{escalate}(x) \iff |C_{\text{cheap}}(x)| \neq 1. \tag{3}$$

A singleton is a statement that, at risk level α, one class is sufficient; anything else is an admission of ambiguity that the cheap sensor cannot resolve. Note what this buys and what it does not. It does *not* promise the singleton is correct — (2) is a marginal statement. It promises that the rate at which we are wrong about being sure is controlled by α, which the integrator chose, and it makes the escalation rate — and therefore the duty cycle of the accelerometer — an observable rather than a tuning outcome.

C. Acquisition cost

We report acquisition cost in operations per window as a hardware-independent proxy: $9N$ operations per channel for the time-domain features and a further $\approx 2.5N \log_2 N + 2N$ when the FFT block is used, with $N = 4096$. This is a proxy and we treat it as one; the quantity that decides a deployment is joules, and mapping ops and sensor duty cycle to joules requires the front-end power figures of a specific node. **[TODO: substituir por joules medidos no ESP32-S3, ou dizer explicitamente que fica para o trabalho de journal]**

III. Experimental setup

Data. We use the KAIST rotating-machinery dataset [9], which provides synchronous vibration (four accelerometers on two housings, 25.6 kHz), three-phase motor current, and two temperature channels, for five conditions — normal, inner-race fault, outer-race fault, misalignment, unbalance — at three torque loads (0, 2, 4 N m). We window at $N = 4096$ samples with 50% hop, keeping 50 windows per recording, giving 2250 windows over 45 recordings. Per channel we extract nine time-domain features (mean, std, RMS, peak-to-peak, kurtosis, skewness, crest, shape and impulse factors) and six spectral features (dominant frequency, centroid, spectral entropy, and energy in three bands). **[TODO: confirmar a taxa de amostragem da corrente no data descriptor — o pipeline atual assume 25.6 kHz para todas as modalidades, o que é conservador contra a nossa própria tese]**

Validation. Windows from one recording are not independent, so every split is at the *recording* level. We report two protocols: (i) GROUPED, a stratified group k-fold on recording id ($k = 3$, capped by the three normal-condition recordings); and (ii) LOLO, leave-one-load-out, training on two torque loads and testing on the third. The first measures generalisation to unseen recordings at a known operating point; the second measures what a deployment actually faces. Calibration sets are carved out of training by holding out whole recordings per class, never windows.

TABLE I
SENSOR-CHAIN FRONTIER. BEST OF {LOGREG, RF, MLP} PER ROW; 95% CI FROM A CLUSTER BOOTSTRAP OVER *recordings*. ACQUISITION COST IS OPERATIONS PER 4096-SAMPLE WINDOW. NINE TIME-DOMAIN FEATURES FROM A *single* ACCELEROMETER LAND WITHIN 0.06 MACRO-F1 OF THE FULL CHAIN AT 18× LOWER COST.

sensor chain	#f	ops/win	grouped	load shift
temperature only	4	4	0.517 [0.44,0.77]	0.299 [0.19,0.40]
current, time only	9	36,864	0.355 [0.30,0.44]	0.167 [0.11,0.23]
1 accel., time only	9	36,864	0.690 [0.66,0.76]	0.687 [0.63,0.74]
4 accel., time only	36	147,456	0.719 [0.68,0.78]	0.744 [0.67,0.79]
current + FFT	15	167,936	0.417 [0.33,0.45]	0.213 [0.13,0.29]
1 accel. + FFT	15	167,936	0.706 [0.68,0.78]	0.714 [0.61,0.80]
current + FFT + temp.	19	167,940	0.587 [0.58,0.75]	0.412 [0.29,0.51]
4 accel. + FFT	60	671,744	0.729 [0.70,0.91]	0.746 [0.68,0.81]
4 accel. + FFT + current	75	839,680	0.729 [0.70,0.92]	0.675 [0.60,0.74]

Models and uncertainty. To make sure the ordering between sensor chains is not an artefact of one architecture, every configuration is run with three classifier families: multinomial logistic regression, a 300-tree random forest, and a two-layer MLP. All reported fits converged. Confidence intervals come from a cluster bootstrap that resamples *recordings* with replacement; a window-level bootstrap gives intervals that are far too narrow, and we regard it as a reporting error. Results are averaged over three seeds.

IV. Results

A. How much sensing is actually needed

Table I gives the cost–accuracy frontier. Two observations matter for a cascade design. Nine time-domain features from a single accelerometer reach 0.687 macro-F1 under load shift against 0.746 for the full four-accelerometer chain with FFT — 18× fewer operations and three fewer conditioning chains for 0.06 macro-F1. The FFT block is similarly close to free to remove. Conversely, current alone degrades from 0.417 to 0.213 under load shift: it is the only configuration that collapses rather than degrades, which bounds how much a current-only cascade stage can be asked to do on this dataset.

Temperature deserves a warning. It reaches 0.517 under grouped validation — better than current with FFT — and 0.181 under load shift. Averaged over a 160 ms window it is near-constant within a recording, so it acts as a recording identifier. Grouped cross-validation does not protect against this; leave-one-load-out exposes it. We suspect this failure mode is common in datasets where slow channels are windowed at the rate of fast ones.

TABLE II

CONFORMAL GATE MINUS SOFTMAX THRESHOLD IN SYSTEM MACRO-F1, COMPARED *at matched escalation rate* (SOFTMAX CURVE LINEARLY INTERPOLATED). THE TWO GATES TRACE THE SAME CURVE; THE CONFORMAL GATE BUYS SPECIFICATION, NOT ACCURACY.

cascade	score	Δ macro-F1	min	max
current $\rightarrow$ 4 accel.+FFT	LAC	-0.0026 ± 0.0122	-0.065	$+0.037$
current $\rightarrow$ 4 accel.+FFT	APS	-0.0025 ± 0.0110	-0.067	$+0.010$
1 accel. $\rightarrow$ 4 accel.+FFT	LAC	-0.0076 ± 0.0253	-0.150	$+0.026$
1 accel. $\rightarrow$ 4 accel.+FFT	APS	$+0.0003 \pm 0.0020$	-0.005	$+0.007$
pooled		-0.0031 ± 0.0153		

TABLE III

COVERAGE GAP (EMPIRICAL $-$ NOMINAL) AFTER RECALIBRATING WITH k RECORDINGS FROM THE TARGET OPERATING POINT. FIVE TARGET RECORDINGS RESTORE NOMINAL COVERAGE — AND *discarding* THE SOURCE CALIBRATION SET BEATS POOLING IT AT EVERY $k \geq 2$.

k	source only	source + target	target only
0	-0.142	–	–
1	-0.142	-0.151	-0.231
2	-0.138	-0.125	-0.073
3	-0.131	-0.099	-0.048
5	-0.134	-0.072	$+0.002$
8	-0.142	-0.037	$+0.012$

B. The gate does not buy accuracy

Table II compares the conformal gate to a swept softmax threshold *at matched escalation rate*, interpolating the softmax curve so the comparison is at equal energy. Pooled over both cascades, both scores, three classifiers and three seeds, the difference is -0.003 ± 0.015 macro-F1. The two gates trace the same curve.

We state this plainly because the alternative framing — selecting the operating points where conformal happens to win — would not survive review, and because the actual contribution does not need it. What changes is the interface: τ is replaced by α, and the escalation rate stops being a tuning outcome.

C. Coverage collapses under operating-point shift

Fig. 1 is the central result. Under grouped validation the guarantee behaves exactly as the theory says: mean gap $+0.046$, conservative, never meaningfully below nominal. Under leave-one-load-out the mean gap is -0.146 and the worst case is -0.453. A nominal 80% interval delivers 55 % empirical coverage.

This is not a surprise — a change of torque load breaks exchangeability between calibration and test, and the guarantee is conditional on exchangeability [10], [11]. It is nonetheless the number a practitioner needs, because the failure is silent: the node keeps emitting singletons at the same rate, and the meaning of those singletons has changed. Panel (b) shows the breakage is also uneven across operating points, so an integrator cannot budget for it with a single safety factor.

D. What it takes to fix it

Table III and Fig. 2 quantify recalibration as a function of the number k of recordings available from the target operating point. Two findings. First, $k = 5$ brings the gap to $+0.002$: recalibration is a commissioning procedure with a number attached, not an open-ended data-collection requirement. Second, and less intuitively, at $k = 3$ recalibrating on target data *alone* gives -0.048 while pooling source and target gives -0.099 — roughly twice as bad. The stale scores drag the quantile in the wrong direction. What the quantile needs is exchangeability with the test point, not sample size, and the natural instinct to keep all available calibration data is actively harmful here.

V. DISCUSSION AND LIMITATIONS

The honest reading of these results is that a conformal escalation gate is a better *interface* than a softmax threshold and not a better *predictor*, and that its guarantee — the thing that makes it worth the trouble — is exactly what breaks first when the machine leaves the operating point it was calibrated at. The practical consequence is that conformal gates should be deployed with a recalibration procedure attached, and we give the size of that procedure.

Limitations. (i) One dataset. The mechanism should be reproduced on at least one further bench — we regard this as the first requirement for a journal version. (ii) 45 recordings, and only three in the normal condition, which caps the number of grouped folds at three and widens every interval. (iii) Cost is reported in operations, not joules; the ranking between cascade designs can change once front-end power is included, and we deliberately do not claim an energy number we have not measured. (iv) Load is the only shift we test; speed, temperature and sensor mounting are all plausible and untested. **[TODO: se conseguires rodar o MAFAULDA antes do prazo, (i) sai daqui e vira uma subseção de resultados — é a mudança que mais aumenta a chance de aceite]**

VI. CONCLUSION

We framed multimodal sensor escalation as a conformal set-size test, replacing an arbitrary confidence threshold with a stated risk level and turning the escalation rate into a measured quantity. On the KAIST bearing dataset the gate matches a tuned softmax threshold in accuracy, and its coverage guarantee — valid under recording-grouped validation — fails under a change of torque load, delivering 55 % where 80% was requested. Five recordings from the target operating point restore the guarantee, and pooling them with the stale calibration set is worse than replacing it. We read this as an argument for treating conformal calibration as a commissioning step tied to the operating point, rather than a one-off performed in the lab.

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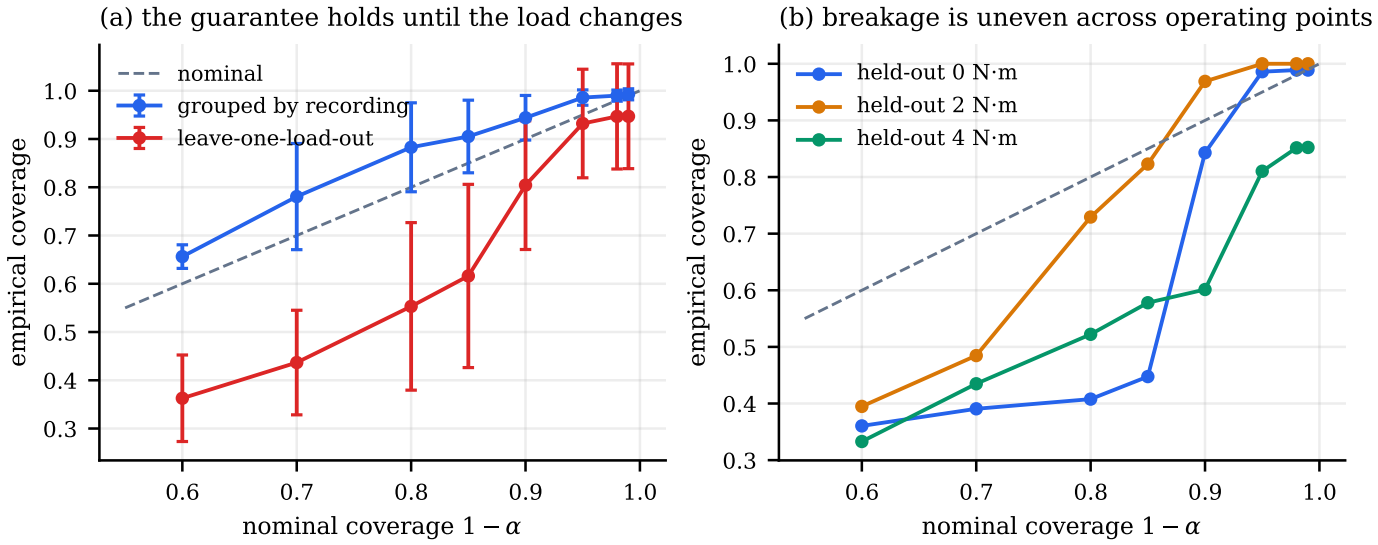


Fig. 1. Empirical versus nominal coverage of the LAC conformal sets. Under recording-grouped validation (a, blue) the guarantee holds and is conservative. Under leave-one-load-out (a, red) it fails: a nominal 80% delivers 55 %. Panel (b) shows the failure is uneven — at $\alpha = 0.20$, coverage is 0.408, 0.729 and 0.522 for held-out 0, 2 and 4 N m. Error bars are one standard deviation over seeds and classifiers.

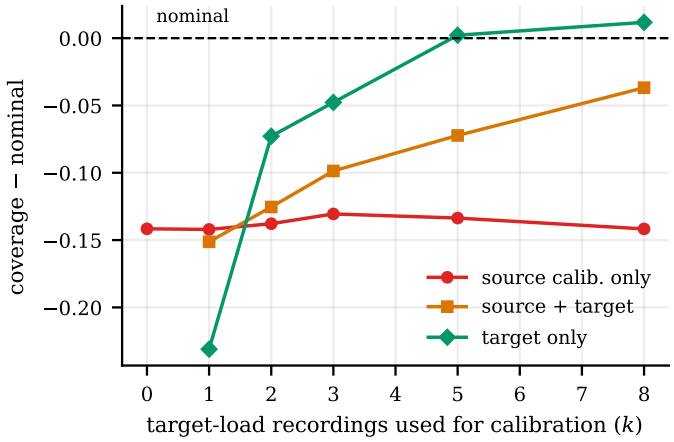


Fig. 2. Recalibration at the target operating point. Five recordings restore nominal coverage. Pooling the stale source calibration set with the new one is consistently worse than discarding it.

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